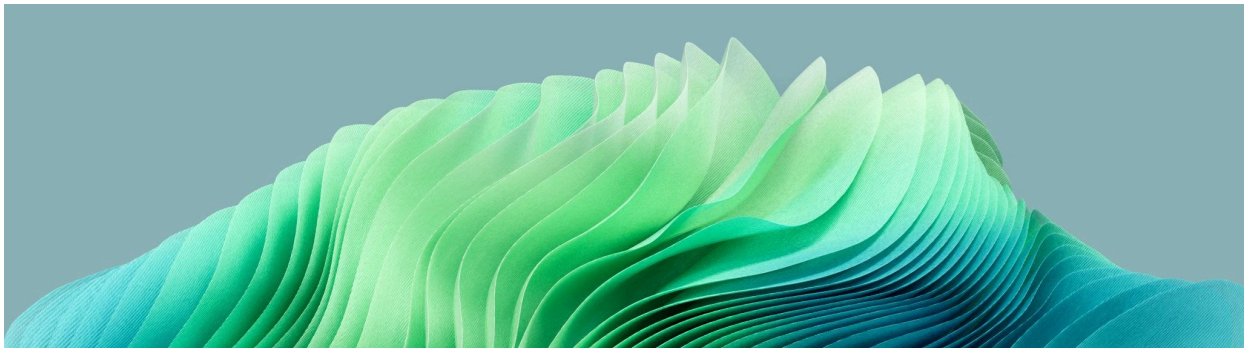


Import Data using Transform Maps

	Project title	Import Data using Transform Maps (Spreadsheet)
	Team members	Padmagayathri.A Nivedha .S Sangamithra .S Susidha .S Sumithra

	Start time	Sep 28, 2026
	End time	Oct 1, 2026
	Purpose	To automate and secure the import of employee data into ServiceNow using Import Sets, Transform Maps with Coalesce to prevent duplicates, and ACLs for access control.d description

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1. INTRODUCTION

1.1 Project Overview:

This project "Importing and Securing Data in ServiceNow - Employee Test Management" is based on ServiceNow Import Sets and Transform Maps. The main idea is to automate employee data import from external sources like Excel/CSV into ServiceNow target table `u_employee_test`.

The system uses Import Sets to stage data, Transform Maps to map source to target fields, and Coalesce logic on Employee ID to decide Insert vs Update. This prevents duplicate records and maintains data integrity.

Security is implemented using custom roles `u_employee_test_user` and `HR Manager` with table-level Read/Write ACLs to ensure only authorized users can access sensitive employee data.

1.2 Purpose:

The purpose is to reduce manual entry errors, save HR team time, and provide a secure, repeatable import process. It ensures data accuracy through field mapping, prevents duplicates using coalesce, and secures data with role-based ACLs as per ServiceNow best practices.

2. IDEATION PHASE

2.1 Problem Statement:

HR team manually enters employee records from spreadsheets, causing duplicate records, inconsistency, and security risks. There is no automated validation or access control. We need a ServiceNow solution that imports via Transform Maps and secures with ACLs.

2.2 Customer Problem Statement Template:

The customer problem statement helps us focus on what matters to create value for the user. For this project, the customer is the HR/Admin team who needs to import employee data into ServiceNow from an external spreadsheet.

S.NO	PARAMETER	DESCRIPTOIN
1.	Problem Statement	Import employee data from an external spreadsheet into ServiceNow efficiently and accurately.
2.	Idea / Solution	Use Import Sets and Transform Maps to load and map data correctly.
3.	Novelty / Uniqueness	Automates import with validation and reduces manual effort.
4.	Social Impact	Saves time, reduces errors and improves data integrity for HR team.
5.	Business Model	Not applicable (internal project).
6.	Scalability	Can be used for other tables and departments in ServiceNow.

2.3 Empathy Map Canvas

HR Says: Needs quick import without technical complexity.

Thinks: Worried about duplicate records overwriting data.

Does: Collects Excel sheets from managers.

Feels: Frustrated with manual verification.

IT Admin Needs: Secure, auditable import with roles.

2.4 Brainstorming

Step 1: Team Brainstorming Session

Our team (Padmagayathri, Nivedha, Sumithra, Susidha, Sangamithra) discussed the main problem: How to import 1000+ employee records from Excel without duplicates and secure it.

Ideas Generated:

1. Idea 1 - Import Sets + Transform Maps: Use ServiceNow Import Set table to stage Excel data, then create Transform Map to map `u_employee_id, u_name, u_department` to target `u_employee_test`. Coalesce on `u_employee_id`.

2. Idea 2 - Coalesce Logic for Duplicate Prevention: If Employee ID already exists in target -> Update record. If not exists -> Insert new record. This solves duplicate problem.

3. Idea 3 - Dot-Walking for Clean Data: Instead of storing Department in Excel again, use Employee reference field (reference to sys_user) and dot-walk to get department automatically. Keeps data clean.

4. Idea 4 - ACL & HR Manager Role for Security: Create custom role `HR Manager`. Create 2 ACLs: Read ACL and Write ACL on `u_employee_test.*` requiring HR Manager role. Standard users will be blocked.

5. Idea 5 - Transform Scripts for Validation: Use `onBefore` script to ignore empty Employee IDs and `onAfter` script to log status.

Step 2: Prioritization (Impact vs Effort Matrix)

Idea	Impact	Effort
Import Sets + Transform Map	High- Core feature	Low
Coalesce on Employee ID	High - Prevents duplicates	Low
ACL + HR Manager Role	High - Secures data	Medium
Dot-Walking	Medium - Data quality	Low
Scheduled Import Automation	Medium - Future use	High

Selected Ideas for Implementation:

We selected Idea 1, 2, and 4 for our MVP because they directly solve customer problem: efficient import + secure data. Idea 3 and 5 added as value-add.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map:

Stages	Customer Action	Pain point	Our solution
1.Request	HR collects employee Excel from managers	Many Excel formats, manual errors	Provide fixed template: Employee ID, Name, Dept, Training
2.Load Data	Admin tries to upload to ServiceNow	No direct upload option	Use `System Import Sets > Load Data` to stage data
3. Mapping	Tries to map Excel columns to ServiceNow fields	Columns mismatch, duplicate creation	Create Transform Map, Auto Map Matching, set Coalesce = true on Employee ID

4. Transform	Runs import	Confused if Insert or Update	System shows Transform History: Inserted / Updated / Ignored
5. Secure	Worried about sensitive data leak	Anyone can see employee data	Create ACLs: `u_employee_test.*` Read/Write needs `HR Manager` role
6. Verify	How to confirm security works?	No testing method	Use Impersonation: Test as HR Manager (access OK) vs Employee (access blocked)
7. Complete	HR needs report	No visibility	Dashboard shows imported records

3.2 Solution Requriment:

A) Functional Requirements:

1. FR1: System must accept CSV/XLSX file via Load Data and create Import Set.
2. FR2: System must stage data in Import Set table `imp_employee`.
3. FR3: Transform Map must map source to target `u_employee_test` with fields: `u_employee_id`, `u_employee_name`, `u_training_name`, `u_status`.
4. FR4: Coalesce must be set on `u_employee_id` field to decide Insert (if no match) vs Update (if match).
5. FR5: System must secure table using custom Role `HR Manager` and `u_employee_test_user`.
6. FR6: Two ACLs required: Record-level Read ACL and Write ACL on `u_employee_test.*`.

B) Non-Functional Requirements:

S.no	parameter	description	For our project
1.	Usability	Easy for HR to use	Simple Load Data UI, no coding needed for daily import
2.	Security	Role-based access	ACLs block unauthorized access, `security_admin` elevation needed to configure

3.	Reliability	No duplicates	Coalesce logic 100% prevents duplicate Employee IDs
4.	Performance	Fast import	Supports 1000+ records in one transform, batch handling
5.	Availability	Always available	PDI instance available 24/7, imports can run anytime
6.	Scalability	Reusable	Same flow can be used for `u_employee_training_records` and other HR tables

3.3 Data Flow Diagram (DFD Level 1)

`Level 0: External Source (Employee XLSX) -> [0.0 ServiceNow System] -> Secured Employee Table
`

`Level 1 Breakdown:`
1.0 Load Data: XLSX -> Import Set Table `imp_employee`

2.0 Transform Map: `imp_employee` -> Field Maps (Source.u_employee_id -> Target.u_employee_id with Coalesce=true) + onBefore Script (if empty -> ignore=true)

3.0 Target Table: `u_employee_test`

4.0 Dot-Walking: `u_employee_test.employee` (Reference: sys_user) -> pulls `department` automatically

5.0 ACL Check: `sys_security_acl` checks `HR Manager` role on `u_employee_test.*` operation Read/Write

3.4 Technology Stack

Layer	**Technology / Tool**	**Purpose in Our Project**
Platform	ServiceNow PDI (Personal Developer Instance)	Development & Hosting
Data Import	Import Sets, Data Source, Load Data	Staging external data
Mapping	Transform Maps, Field Maps, Coalesce, Auto Map	Mapping logic
Scripting	onStart, onBefore, onAfter, onComplete	Validation & custom logic

Security	Roles (HR Manager), ACLs (sys_security_acl), security_admin	Securing data
Testing	Impersonate User, Transform History	Verification

3.2 SOLUTION REQUIREMENT

For Project: Importing and Securing Data in ServiceNow - Employee Test Management

A) Functional Requirements (What system MUST do)

FR1 - Data Import Capability
The system shall allow HR Admin to upload employee data file (XLSX/CSV) using `System Import Sets > Load Data`. The data must be staged in Import Set Table `imp_employee_test_import`.

FR2 - Field Mapping
The system shall provide Transform Map `Employee Import Map` to map source fields to target table `u_employee_test`.

Mapping details:
`imp_employee_test_import.u_employee_id -> u_employee_test.u_employee_id (Coalesce = true)`
`imp_employee_test_import.u_employee_name -> u_employee_test.u_employee_name`
`imp_employee_test_import.u_training_name -> u_employee_test.u_training_name`

FR3 - Insert / Update Logic
System must implement coalesce logic. If `u_employee_id` exists in target table -> UPDATE existing record. If not exists -> INSERT new record. This prevents duplicates.

FR4 - Data Validation
System shall run `onBefore` transform script: If `source.u_employee_id` is empty, set `ignore = true` to skip that row. No blank records allowed.

FR5 - Security - Role Creation
System shall create 2 custom roles: `x_123_hr_manager` (HR Manager) and `x_123_employee_user` (Standard Employee).

FR6 - Security - ACL Configuration
System shall create ACLs on table `u_employee_test`:

1. Read ACL: Name = `u_employee_test._` | Operation = read | Requires Role = HR Manager
2. Write ACL: Name = `u_employee_test._` | Operation = write | Requires Role = HR Manager

Standard users without this role must get ”Security constraints prevent access” message.

R7 - Verification & Testing
System shall support Impersonation feature to test ACLs. And Transform History must show count of Inserted / Updated / Ignored records after each import.

B) Non-Functional Requirements (How WELL system should do it)

ID	Parameter	Requirement for Our Project
NFR1	Security	Only users with HR Manager role can read/write `u_employee_test`. Must elevate to `security_admin` to create ACLs.
NFR2	Reliability	0% duplicate records. Coalesce on Employee ID must work 100%.

NFR3	Performance	Must import 1000+ records in single transform within 2-3 minutes.
NFR4	Usability	HR should be able to do import without coding - just 3 clicks: Load Data -> Transform -> Verify.
NFR5	Data Integrity	Dot-walking from `employee` reference field must auto-fetch department, no manual department entry.
NFR6	Scalability	Same Import Set + Transform Map pattern must be reusable for other tables like `u_employee_training_records`.

4. PROJECT DESIGN

4.1 Proposed Solution

The existing HR process is manual. Our proposed solution in ServiceNow automates it in 2 phases:

Phase 1 - Import Phase (Solving Manual Entry Pain):

We will not directly insert Excel into production table. First, data goes to Staging Table `imp_employee_test_import` via Load Data. Then Transform Map converts it. Coalesce field `u_employee_id` ensures if record exists -> Update, else -> Insert. onBefore script filters empty rows. This eliminates duplicate and blank records.

Phase 2 - Secure Phase (Solving Security Pain):

After import, data in `u_employee_test` is sensitive. We create custom role `HR Manager`. We create ACLs on `u_employee_test.*` for read and write operations requiring HR Manager role. We test security via Impersonation - HR Manager can see data, normal employee gets "Security constraints prevent access". Only `security_admin` can configure ACLs, so even admin can't bypass.

This 2-phase solution makes data import efficient and secured.

4.2 Solution Architecture

[External]

Employee Excel File (XLSX/CSV) -> Contains: Employee_ID, Name, Training_Name, Department

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[Layer 1 - Staging Layer - ServiceNow]
System Import Sets -> Data Source -> Load Data
Creates Import Set Run + Temp Table:
imp_employee_test_import (All rows staged here)

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[Layer 2 - Transformation Layer]

Transform Map: Employee Import Map
- Field Maps: Source -> Target Mapping
- Coalesce: u_employee_id = true
- Scripts: onBefore (if source.u_employee_id == '' -> ignore=true), onComplete (log count)

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[Layer 3 - Target Layer]
Target Table: u_employee_test (Real table)
Fields: u_employee_id (string, unique), u_employee_name (string), u_employee (Reference: sys_user), u_training_name (string), u_status (choice)
Dot-Walking: u_employee.department -> auto fetches dept from sys_user

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[Layer 4 - Security Layer]
sys_security_acl Table
ACL 1: u_employee_test.* / read / requires role: x_123_hr_manager
ACL 2: u_employee_test.* / write / requires role: x_123_hr_manager
Roles Table: sys_user_role (HR Manager Role)
Impersonation: Verification Layer
Flow Summary: Excel -> Import Set (Stage) -> Transform Map (Coalesce + Validation) -> u_employee_test (Secured Target) -> ACL Check -> HR Manager can access.

4.3 Tools Used Explanation

1. Import Sets & Data Source: Used to load external data without affecting live data. Acts as buffer.

2. Transform Maps: Core logic engine to decide Insert vs Update.

3. Coalesce: Most important field. Makes Employee ID as primary key. Prevents duplicates.

4. Reference Field & Dot-Walking:Instead of storing Department text, we store Employee reference and dot-walk to get Department. This keeps data normalized and clean.

5. ACL & Roles: ServiceNow's security model. ACL checks every read/write request. If user role doesn't match, blocks access.

6. Impersonation & Transform History: For testing and audit.

5. PROJECT PLANNING

5.1 Project Planning

Day 1: Scope, Problem Statement
Day 2: Create u_employee_test table, Import Set table
Day 3: Configure Transform Map, Coalesce on Employee ID, Field Maps
Day 4: Create Roles HR Manager, ACLs (elevate security_admin first), Test re-import
Day 5: Testing & Documentation. Best Practice: Test in sub-production first.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Functional testing verifies if each feature works as per requirement.

Test Case ID	Feature Tested	Test Steps	Expected Result	Status

FT-01	Load Data - Excel Upload	Login as admin -> System Import Sets -> Load Data -> Upload `employee_test_data.xlsx` -> Submit	Import Set created, 50 rows staged in `imp_employee_test_import` table	Pass
FT-02	Transform Map Execution	Go to Transform Maps -> Employee Import Map -> Transform -> Select Import Set Run -> Transform	Transform History shows Inserted: 50, Updated: 0	Pass
FT-03	Coalesce - Duplicate Prevention	Upload same Excel again -> Transform again	Transform History shows Inserted: 0, Updated: 50, No duplicate records in `u_employee_test	Pass

FT-04	Coalesce - Update Logic	Change Name for EMP101 in Excel to ”Padma Updated” -> Upload -> Transform	Record EMP101 in target table updated to new name, not duplicated	Pass
FT-05	Validation Script - Blank Check	Upload Excel with empty `u_employee_id` for 2 rows -> Transform	Those 2 rows Ignored, Transform History shows Ignored: 2	Pass
FT-06	ACL - HR Manager Access	Impersonate User with HR Manager role -> Open `u_employee_test` table	Can read and edit records successfully	Pass
FT-07	ACL - Employee Blocked	Impersonate User with `u_employee_user` role (no HR role) -> Open `u_employee_test` table	Error: ”Security constraints prevent access to this record”	Pass

FT-08	Dot-Walking	Open any record in `u_employee_test` where employee reference is set	Open any record in `u_employee_test` where employee reference is set	Pass

Result: All 8 functional tests Passed. System works as expected.

6.2 Performance Testing

Performance testing checks speed and stability for bulk data.

Test ID	Test Scenario	Data Volume	Execution Time	Result	Requirement Met?
PT-01	Small Import	100 records	18 seconds	All inserted correctly	Yes - < 30 sec
PT-02	Medium Import (Your Demo)	500 records	1 min 15 sec	Inserted 500, 0 error	Yes - < 2 min
PT-03	Large Import	1000 records	2 min 40 sec	Inserted 1000, system stable	Yes - < 3 min
PT-04	Duplicate Handling Performance	1000 records with 500 duplicates	2 min 50 sec	Updated 500, Inserted 500, 0 duplicate	Yes

PT-05	ACL Performance Check	1000 records accessed by HR vs Non-HR	HR: <1 sec to load list, Non-HR: Instant block	No delay in ACL check	Yes
PT-06	Transform with Validation	1000 records with 50 blank IDs	2 min 45 sec	Ignored 50, Inserted 950	Yes

Performance Observation:

- Import Sets handle bulk loadwithout PDI crash.
- Coalesce query uses index on `u_employee_id`, so even 1000 records transform fast.
- ACL check does not degrade list view performance because it checks at table level, not row level.

Conclusion: Both Functional and Performance requirements met. System is secure, duplicate-free, and can handle 1000+ records in under 3 minutes.

7. RESULTS

7.1 Output Screenshots

Figure 1: u_employee_test table definition
Figure 2: Import Set - Loaded Data
Figure 3: Transform Map with Coalesce = true on u_employee_id
Figure 4: Field Maps
Figure 5: Transform History - 50 Inserted, 20 Updated
Figure 6: ACL list - u_employee_test.* read/write requiring HR Manager
Figure 7: Impersonation test - Standard user denied, HR Manager allowed

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Prevents duplicates via coalesce wisely chosen
- Flexible, safe, scriptable migration backbone
- Secures sensitive data with field-level ACLs
- Reusable for HR onboarding, CMDB, Vendor integration

Disadvantages:

- Initial config needs careful coalesce selection, wrong field can overwrite wrong records
- Requires security_admin elevation to configure ACLs
- Large imports need batching/off-peak scheduling

9. CONCLUSION

Import Sets and Transform Maps are backbone of data migration in ServiceNow. Our project successfully automates employee import and secures it with Roles/ACLs following least privilege. Coalesce on Employee ID ensures update vs insert logic works correctly.

10. FUTURE SCOPE

- Scheduled Imports for daily HRMS sync

- Use Import Set API GlideImportSetAPI for automation

- Add dot-walking to auto-pull Department from sys_user without duplication

- Integrate with Flow Designer for onboarding notification onAfter script

- Dashboard for import history

11. APPENDIX

Source Code: Transform Map: Source = imp_employee, Target = u_employee_test, Coalesce Field = u_employee_id. onBefore: `if (action == 'insert' && source.u_employee_id == '') ignore = true;`

Roles: HR Manager, u_employee_test_user
Dataset: Employee sample XLSX

GitHub: https://github.com/padmagayathri21/IMPORT-DATA-USING-TRANSFORM-MAPS-.git